

MA 321 (Optimization)
End-semester Examination
20th November, 2024

Time: 2 pm - 5 pm

Maxim marks: 40

Let (P) be Minimize $f(\mathbf{x})$, subject to $g_i(\mathbf{x}) \leq 0$, for $i = 1, \dots, m$.

Notations: For $\mathbf{x}^* \in \text{Fea}(\mathbf{P})$, $D_{\mathbf{x}^*}$ is the set of all feasible directions at $\mathbf{x}^*$.

$I_{\mathbf{x}^*} = \{i \in \{1, \dots, m\} : g_i(\mathbf{x}^*) = 0\}$, $G_0(\mathbf{x}^*) = \{\mathbf{d} \in \mathbb{R}^n : \nabla g_i(\mathbf{x}^*)\mathbf{d} < 0 \text{ for all } i \in I_{\mathbf{x}^*}\}$.

No credit will be given for answers without justification or wrong justification

1. Consider the following problem (P):

Minimize $f(x_1, x_2) = x_1^2 x_2 - x_1 + 2x_2 - x_1 x_2$

subject to $x_1^2 + x_2^2 \leq 1$,

$2x_1 + x_2 \leq 2$,

$-x_1 \leq 0, -x_2 \leq 0$.

- (a) Find all $\mathbf{x}^* \in \text{Fea}(\mathbf{P})$ on the boundary of $\text{Fea}(\mathbf{P})$ for which $G_0(\mathbf{x}^*) \neq D_{\mathbf{x}^*}$.
- (b) Check whether $\mathbf{x}^* = [0, 1]^T$ satisfies the first order necessary conditions for a local minimum of (P). If yes justify, if no then give a feasible direction $\mathbf{d}$ at $\mathbf{x}^*$, such that $f(\mathbf{x}^* + t\mathbf{d}) < f(\mathbf{x}^*)$ for $t > 0$, sufficiently small.
- (c) Is $\mathbf{x}^* = [\frac{3}{5}, \frac{4}{5}]^T$ an FJ point of (P)?
- (d) Are all the FJ points of (P) also KKT points?
- (e) Does $\mathbf{x}^* = [1, 0]^T$ satisfy the second order necessary conditions for a local minimum?

[2 + 2 + 3 + 2 + 3]

2. Consider the following problem (P):

Minimize $ex_1^2 + 2x_2^2 - x_1 x_2 - x_1 + 3x_2$

subject to $ax_1 + bx_2 \leq 2$,

$cx_1 + dx_2 \geq 4$,

$-x_1 \leq 0, -x_2 \leq 0$,

such that each of a, b, c, d, e are nonzero.

- (a) If possible give values of a, b, c, d, e , such that every KKT point (there exists atleast one) of (P) is a global minimum.
- (b) If possible give values of a, b, c, d, e and an $\mathbf{x}^* \in \text{Fea}(\mathbf{P})$ such that there exists an FJ point which is not a local minimum point.

[3 + 3]

All parts in the above question are independent

3. Using the Big(M) method find an initial BFS of (P) given below:

Minimize $-x_1 + 4x_2 + 2x_3$

subject to $2x_1 + x_2 - 3x_3 \leq 2$,

$x_1 - x_2 + 4x_3 \geq 4$,

$x_1 \geq 0, x_2 \geq 0, x_3 \geq 0$.

[3]

4. Consider the following transportation problem (P) with cost c_{ij} 's (given in the cells), with supply s_i 's (20,30,40) and demand d_j 's (30,35,10,15) given below:

4	7	1	5	20
9	8	13	7	30
4	5	9	3	40
30	35	10	15	

- (a) Check whether the initial BFS x_0 with basic cells $B = \{(1, 1), (2, 2), (3, 1), (3, 2), (3, 3), (3, 4)\}$, is optimal for (P) (by taking $u_3 = 0$, where u_3 is the dual variable corresponding to supply s_3). If it is not optimal, then by using a θ -loop, get a BFS with a lower value of the objective function.
- (b) If cost c_{ij} is changed to c'_{ij} for some x_{ij} such that there exists an optimal solution of new (P) with $x_{11} = 17$, then does this new (P) have a unique optimal solution? [4 + 2]

5. Let B be a basic set of cells in a 5×5 transportation array and let θ_1, θ_2 be the θ -loops in $B \cup \{(p, q)\}$, $B \cup \{(r, s)\}$, respectively, such that $\theta_1 \cap \theta_2 = \emptyset$. Let $(u, v) \in \theta_1$.

(a) Is $B \cup \{(p, q)\} \setminus \{(u, v)\}$ LI?

(b) What are the possible number of distinct θ -loops in $B \cup \{(p, q), (r, s)\}$? [5]

6. Consider a linear programming problem (P) of the form,

Minimize $\mathbf{c}^T \mathbf{x}$

subject to $\mathbf{A}_{m \times n} \mathbf{x} = \mathbf{b}$, $\mathbf{x} \geq 0$.

Let LP(M) be obtained from (P) by adding atleast one artificial variable to (P) with cost M (M large). Let x_0 be an optimal BFS for (P) and let y_0 be optimal for the dual of (P).

Check the correctness of the following statements with proper justification.

- (a) If $\text{Fea}(\mathbf{P})$ is unbounded then $\text{Fea}(\text{LP}(\mathbf{M}))$ is unbounded.
- (b) If $\text{Fea}(\text{LP}(\mathbf{M}))$ is unbounded then $\text{Fea}(\mathbf{P})$ is unbounded.
- (c) If a variable x_{n+1} is added to (P) and y_0 satisfies the newly added constraint to the dual of new (P) as an equality then y_0 is optimal for the dual of new (P).
- (d) If c_1 is changed to c'_1 such that the optimal value of the new (P) is strictly greater than $\mathbf{c}^T x_0$ then x_1 must be a basic variable in BFS x_0 . [10]

All parts in the above question are independent

*** END ***